

INTERNSHIP REPORT
On
ARTIFICIAL INTELLIGENCE INTERNSHIP

**Submitted
To**



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CANDIDATE'S DECLARATION

I, SUMIT KUMAR, hereby submit this internship report titled “Artificial Intelligence Internship” in partial fulfilment of the academic requirements of the B.Tech CSE with Specialization in AI/ML programme at IILM University, Greater Noida.

This report has been prepared as an academic document based on the internship programme materials available to me and the project themes specified for the Artificial Intelligence internship. The technical project sections are presented as structured project studies/prototypes and are not intended to represent independently verified industry production results.

I have used references wherever external concepts, libraries, datasets, or documentation are discussed. I understand that the accuracy and originality of the academic material submitted in this report are my responsibility.

A handwritten signature in black ink that reads "Sumit Kumar". The signature is written in a cursive style and is positioned above the printed student name.

Student Name: SUMIT KUMAR

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing an academic environment in which I could explore the practical dimensions of Artificial Intelligence and Machine Learning.

I am thankful to the internship programme team at Codec Technologies for providing an internship framework focused on Artificial Intelligence and for defining project themes that connect fundamental concepts with practical applications. The programme material covers areas such as Python, machine learning, neural networks, natural language processing, and applied AI projects.

I am also grateful to my faculty members for their academic guidance and to my parents and well-wishers for their continuous encouragement. Their support helped me approach the subject with discipline and curiosity.

Finally, I appreciate the opportunity to document the concepts, system designs, algorithms, implementation approach, and expected outcomes associated with the selected AI projects in a structured academic format.



Student Name: SUMIT KUMAR

Roll No.: 25SCS1003003501

INTERNSHIP COMPLETION CERTIFICATE



Figure 3.1: Internship Certificate

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4. PROJECT DESCRIPTION

4.1 INTRODUCTION

Artificial Intelligence (AI) is a major area of computer science concerned with building systems that can perform tasks requiring learning, pattern recognition, prediction, classification, language understanding, and decision support. Machine Learning (ML), a core part of AI, allows computational models to learn useful relationships from data rather than relying only on manually written rules.

The internship theme provided for the programme is Artificial Intelligence. The project-work section associated with the programme lists several application-oriented project choices, including Stock Price Predictor, Sentiment Analysis on Twitter Data, Handwritten Digit Recognizer, Movie Recommendation System, Customer Churn Prediction, Chatbot for Customer Service, Spam Email Classifier, and Image Classifier for Fruits.

For this report, two project themes have been selected because they demonstrate two complementary AI capabilities: numerical prediction and natural-language classification. The first project is a Stock Price Predictor based on historical numerical data. The second is a Sentiment Analysis system that classifies text into positive, negative, or neutral sentiment.

The purpose of this report is to document a professional project design, implementation plan, algorithmic reasoning, technology stack, system architecture, methodology, and expected outcomes. The presentation is intentionally structured to show how an AI project can move from problem definition and data preparation to model development, evaluation, and responsible interpretation.

4.1 INTRODUCTION — CONTINUED

AI projects are most effective when the problem is clearly defined before model selection. A technically advanced model cannot compensate for poor data quality, unclear target variables, data leakage, inappropriate evaluation metrics, or unrealistic expectations. Therefore, both projects in this report follow a disciplined lifecycle: define the objective, understand the data, preprocess the inputs, engineer useful features, train a suitable model, evaluate it with appropriate metrics, and document limitations.

The Stock Price Predictor is treated as a supervised regression problem. Historical observations can contain variables such as opening price, highest price, lowest price, closing price, trading volume, and lagged values. A regression model can learn relationships between selected historical features and a target value such as the next closing price. Because financial markets are affected by many external factors, the output should be understood as a model estimate rather than a guaranteed future price.

The Sentiment Analysis project is treated as a supervised text-classification problem. Text is cleaned and transformed into numerical representations using Natural Language Processing (NLP) techniques. A classifier can then learn patterns associated with positive, negative, and neutral language. This type of system can be applied to large volumes of customer feedback, social-media text, reviews, or survey comments.

Together, the two projects provide a balanced demonstration of AI concepts relevant to a CSE student specializing in AI/ML: regression, classification, preprocessing, feature engineering, model evaluation, NLP, visualization, and practical system design.

Figure 4.1: Overall AI Internship Project Workflow

Stage	Purpose
1. Problem Definition	Define the prediction/classification target and constraints.
2. Data Preparation	Collect, clean, validate, and structure the input data.
3. Feature Engineering	Create meaningful numerical or textual representations.
4. Model Development	Train and tune an appropriate ML model.
5. Evaluation	Measure performance using suitable validation metrics.
6. Interpretation	Explain outputs, limitations, and responsible use.

4.2 ORGANIZATION PROFILE

According to the internship offer letter, Codec Technologies presents itself as a global platform focused on empowering learners and connecting diverse talent to create career opportunities worldwide. The same document states that Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries.

The offer letter identifies the internship as a one-month programme for the designation Artificial Intelligence Intern, with a Pan India, Hybrid mode. The stated duration is 01/08/2026 to 01/09/2026, with reporting to assigned project head(s).

Particular	Details
Organization	Codec Technologies Pvt. Ltd.
Internship Designation	Artificial Intelligence Intern
Programme Type	1 Month Internship
Mode / Location	Pan India, Hybrid
Duration	01/08/2026 – 01/09/2026
Reporting	Assigned Project Head(s)
Academic Area	Artificial Intelligence / Machine Learning
Offer Letter Date	01/08/2026

The programme communication indicates that interns are expected to complete assigned projects and training tasks through the platform and email. It also mentions weekly performance reviews and the possibility of a scholarship award based on performance. These programme features establish a structure in which theoretical AI concepts can be connected to application-oriented project themes.

4.3 PROBLEM STATEMENT

Project 1 — Stock Price Predictor

Financial time-series data contains sequential observations whose values may be influenced by historical patterns, market conditions, trading activity, and external information. Manually examining large historical datasets is difficult and does not provide a consistent quantitative prediction mechanism. The project therefore defines a supervised regression problem in which historical market variables are transformed into model features and a regression algorithm estimates a selected future price value.

The key challenge is not simply producing a number; it is constructing a reliable data pipeline that avoids leakage, preserves chronological order, handles missing values, and evaluates the model on data that was not used for training.

Project 2 — Sentiment Analysis on Twitter Data

Large volumes of short text can contain opinions, reactions, and attitudes. Reading and categorizing each message manually is slow and difficult to scale. The objective of this project is to design an NLP classification pipeline that accepts text and assigns one of three sentiment classes: positive, negative, or neutral.

The challenge is that language is noisy and context-dependent. Informal spelling, abbreviations, punctuation, emojis, hashtags, negation, and ambiguous expressions can affect classification. Therefore, preprocessing and appropriate feature representation are essential parts of the system.

Combined Problem Perspective

The two projects demonstrate how the same AI development lifecycle can be adapted to different data modalities: structured numerical time-series data for regression and unstructured text data for classification.

4.4 PROJECT OBJECTIVES

Objective	Description
O1	Understand the end-to-end workflow of an applied AI/ML project.
O2	Prepare structured historical data for a regression model.
O3	Develop a logical feature set for stock-price estimation.
O4	Apply suitable regression metrics such as MAE, MSE/RMSE and R^2 .
O5	Build an NLP pipeline for text cleaning and representation.
O6	Classify text into positive, negative, and neutral sentiment.
O7	Use an appropriate classification model and evaluation metrics.
O8	Visualize data patterns and model evaluation information.
O9	Document assumptions, limitations, risks, and possible improvements.
O10	Demonstrate professional AI project documentation and reproducibility.

Learning Objectives

- Strengthen Python programming and data-handling skills.
- Understand the role of NumPy and Pandas in data processing.
- Use Matplotlib for exploratory visualization.
- Understand the difference between regression and classification.
- Apply Scikit-learn concepts such as train/test splitting, pipelines, preprocessing, and evaluation.
- Understand why model evaluation must match the type of problem.
- Develop the habit of documenting limitations instead of overclaiming model performance.

4.5 SCOPE OF THE PROJECT

Scope of Stock Price Predictor

- Import historical price data from a CSV or equivalent structured source.
- Validate column types, missing values, duplicate records, and chronological ordering.
- Generate lag-based and rolling statistical features where appropriate.
- Split data chronologically into training and evaluation portions.
- Train a baseline regression model and compare it with an optional advanced model.
- Evaluate prediction error using MAE, RMSE and R^2 .
- Visualize actual versus predicted values for interpretation.

Scope of Sentiment Analysis

- Accept a labeled text dataset containing short messages or social-media posts.
- Clean text by handling URLs, mentions, unnecessary whitespace, and selected punctuation.
- Convert text to numerical features using TF-IDF or another suitable representation.
- Train a multiclass classifier for positive, negative, and neutral labels.
- Evaluate using accuracy, precision, recall, F1-score and a confusion matrix.
- Provide a prediction function for new text inputs.

Out of Scope

- Real-time financial trading or automated investment decisions.
- Guaranteed stock-price forecasting or financial advice.
- Production-scale social-media data collection requiring platform-specific access.
- Claims of live deployment, industry-level accuracy, or independently verified internship deliverables not documented in the supplied materials.

4.6 TECHNOLOGIES AND TOOLS USED

Technology / Tool	Role in Project
Python	Primary programming language for AI/ML implementation.
NumPy	Numerical arrays, mathematical operations, and efficient computation.
Pandas	Data loading, cleaning, transformation, and tabular analysis.
Matplotlib	Exploratory data visualization and model-performance plots.
Scikit-learn	Preprocessing, model training, evaluation, and ML utilities.
Jupyter Notebook	Interactive development, experimentation, and documentation.
VS Code	Source-code editing and project organization.
CSV / Tabular Data	Convenient storage format for structured datasets.
NLP / TF-IDF	Text normalization and conversion into machine-readable features.

Development Environment

A practical implementation can be organized as a small Python project containing separate modules for data loading, preprocessing, model training, evaluation, visualization, and prediction. Keeping these responsibilities separate improves maintainability and makes it easier to replace a model without rewriting the complete pipeline.

Functional Requirements

ID	Requirement
FR1	System shall accept structured historical data for stock prediction.
FR2	System shall validate and preprocess input data.
FR3	System shall train a regression model and generate estimates.
FR4	System shall accept text input for sentiment classification.
FR5	System shall preprocess text consistently during training and inference.
FR6	System shall return one of the defined sentiment classes.
FR7	System shall report evaluation metrics for model assessment.

4.7 SYSTEM ARCHITECTURE

4.7.1 Overall Architecture

The architecture is divided into data ingestion, preprocessing, feature engineering, model development, evaluation, and inference. This layered approach keeps the system understandable and reduces the risk of applying training-time transformations differently from inference-time transformations.

Layer	Stock Predictor	Sentiment Analysis
Input	Historical market records	Labeled text records
Validation	Schema, missing values, order	Text presence, labels, duplicates
Preprocessing	Scaling/cleaning as required	Normalization, token/text cleaning
Features	Lag/rolling/statistical variables	TF-IDF or equivalent text vectors
Model	Regression algorithm	Multiclass classifier
Evaluation	MAE, RMSE, R^2	Accuracy, Precision, Recall, F1
Inference	Estimated future value	Sentiment class

4.7.2 Data Flow

Data flows from a raw input source to a validated dataset. The preprocessing layer converts the raw data into a consistent representation. The feature-engineering layer constructs model-ready variables. The trained model produces an output, which is then interpreted through evaluation metrics and visualizations. The same preprocessing logic should be reused during inference to maintain consistency.

4.7 SYSTEM ARCHITECTURE — CONTINUED

4.7.3 Stock Price Predictor Architecture

Component	Description
Data Source	Historical OHLCV-style records or equivalent structured price data.
Data Validator	Checks date order, null values, duplicates, and numeric columns.
Feature Builder	Creates lag values, rolling statistics, and selected market features.
Train/Test Splitter	Separates earlier observations for training and later observations for evaluation.
Regression Model	Learns the relationship between features and target price.
Evaluator	Computes MAE, RMSE and R^2 .
Visualizer	Plots trends and actual-versus-estimated values.

Figure 4.2: Stock Price Predictor Architecture

4.7.4 Sentiment Analysis Architecture

Component	Description
Text Input	Labeled short messages for supervised learning.
Cleaner	Normalizes text and removes selected noise.
Vectorizer	Transforms text into numerical features, e.g., TF-IDF.
Classifier	Predicts positive, negative, or neutral class.
Evaluator	Produces classification metrics and confusion matrix.
Inference Module	Accepts new text and returns the predicted class.

Figure 4.3: Sentiment Analysis Architecture

4.8 METHODOLOGY

The methodology follows a structured machine-learning lifecycle. The central principle is reproducibility: every transformation applied to training data should be explicitly documented and, where possible, implemented as part of a reusable pipeline.

Phase 1 — Requirement Analysis

The first phase identifies the prediction target, input variables, expected output, constraints, and evaluation criteria. For the stock project, the target is a future price-related value. For the sentiment project, the target is one of three sentiment classes.

Phase 2 — Data Understanding

Data understanding includes checking the shape of the dataset, data types, missing values, duplicate records, label distribution, and unusual values. Visual inspection is useful because numerical summaries alone may not reveal trends or imbalance.

Phase 3 — Data Preprocessing

Preprocessing converts raw inputs into a consistent form. Numerical data may require missing-value handling, scaling, and date parsing. Text data requires normalization and cleaning. Preprocessing decisions should be justified by the data rather than applied mechanically.

Phase 4 — Feature Engineering

Features are the measurable representations used by the model. For stock prediction, lagged prices and rolling statistics can provide historical context. For sentiment analysis, TF-IDF represents how important words or n-grams are within a corpus while reducing the influence of extremely common terms.

4.8 METHODOLOGY — STOCK PRICE PREDICTOR

Step 1: Data Preparation

The dataset is expected to contain a chronological date field and numerical market variables. The date is parsed into a proper date type and the records are sorted chronologically. Invalid numeric values and missing observations are handled according to a documented rule.

Step 2: Feature Engineering

Feature	Purpose
Previous Close (Lag 1)	Provides the immediately preceding closing value.
Lag 2 / Lag 3	Adds short historical context.
Rolling Mean	Smooths short-term fluctuations.
Rolling Standard Deviation	Represents recent variability.
Volume	Provides a measure of trading activity where available.
High-Low Range	Represents the intraday price range.

Step 3: Model Development

A linear regression model can be used as a transparent baseline because its coefficients are relatively easy to interpret. A more advanced regression method may be evaluated if the baseline is insufficient. The model should be trained only on the training period, while later chronological observations are reserved for evaluation.

Step 4: Evaluation

Mean Absolute Error (MAE) measures the average absolute difference between estimated and actual values. Root Mean Squared Error (RMSE) gives greater weight to larger errors. R^2 indicates how much of the variance in the target is explained by the model relative to a baseline. No single metric should be treated as a complete description of forecasting quality.

4.8 METHODOLOGY — SENTIMENT ANALYSIS

Step 1: Text Cleaning

- Convert text to a consistent case where appropriate.
- Remove or normalize URLs and user mentions.
- Normalize repeated whitespace and unnecessary symbols.
- Handle hashtags according to the chosen preprocessing policy.
- Preserve meaningful negation/context where it can affect sentiment.

Step 2: Text Representation

TF-IDF is a practical baseline for short-text sentiment classification. It converts documents into sparse numerical vectors. Terms that are frequent in a particular document but less common across the corpus receive higher importance than terms that occur almost everywhere.

Step 3: Classification

A multiclass classifier such as Logistic Regression or another Scikit-learn classification model can learn the mapping from TF-IDF features to sentiment labels. A pipeline is recommended so that vectorization and classification are applied consistently.

Step 4: Evaluation

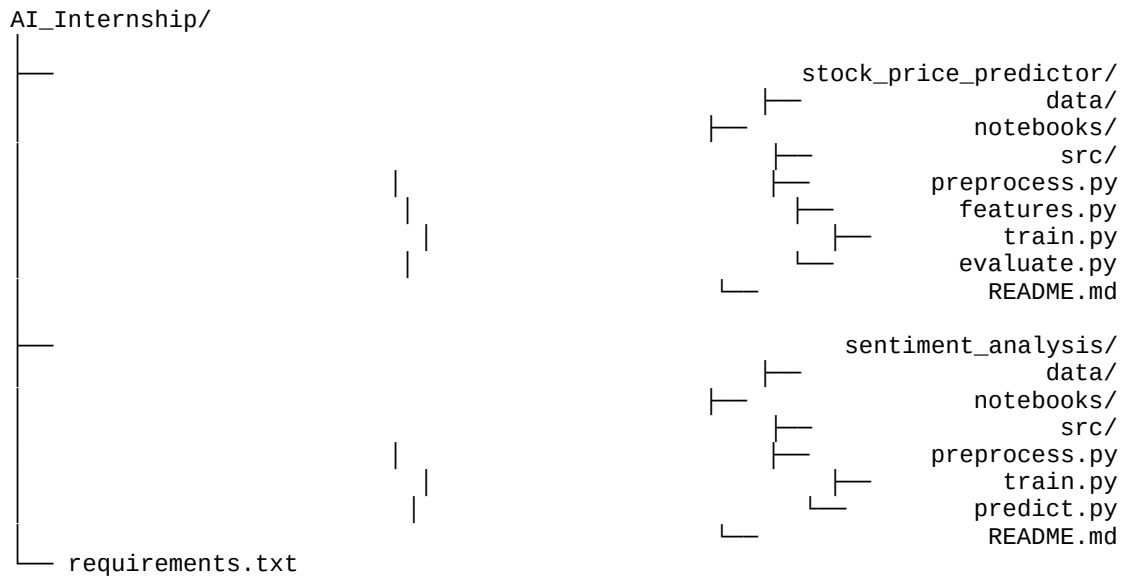
Metric	Meaning
Accuracy	Fraction of predictions that are correct overall.
Precision	How often predictions for a class are correct.
Recall	How many actual examples of a class are identified.
F1-score	Harmonic balance of precision and recall.
Confusion Matrix	Shows class-wise correct and incorrect predictions.

Sentiment Classes

Class	Interpretation
Positive	Text expresses a favorable, approving, or optimistic attitude.
Negative	Text expresses an unfavorable, critical, or pessimistic attitude.
Neutral	Text is informational, balanced, or does not show a clear polarity.

4.8 METHODOLOGY — IMPLEMENTATION DESIGN

Recommended Project Structure



Stock Model Pseudocode

```
load historical data
sort records by date
validate numeric fields
create lag and rolling features
remove rows made incomplete by lagging
split data chronologically
fit regression model on training period
predict evaluation on period
calculate MAE, RMSE and R2
plot actual vs predicted values
```

Sentiment Model Pseudocode

```
load labeled text data
clean and normalize text
split into training and evaluation sets
fit TF-IDF vectorizer on training text only
transform training and evaluation text
fit multiclass classifier
predict evaluation labels
calculate accuracy, precision, recall and F1
inspect confusion matrix
save pipeline for consistent inference
```

4.8 METHODOLOGY — VALIDATION, SECURITY AND RESPONSIBLE AI

Validation Strategy

Validation must reflect the nature of the data. Time-series observations should be evaluated chronologically rather than randomly mixing future observations into the training set. For text classification, a stratified split can be used when appropriate so that the class distribution is represented across training and evaluation sets.

Avoiding Data Leakage

Data leakage occurs when information that would not be available at prediction time is accidentally used during training. Examples include calculating a feature with future observations or fitting a text vectorizer on the complete dataset before the train/test split. Using a proper pipeline and fitting transformations only on training data helps prevent this issue.

Model Limitations

- Stock prices are influenced by many factors that may not be present in historical price features.
- Short-text sentiment can be ambiguous and may require contextual understanding.
- Class imbalance can make accuracy misleading in sentiment classification.
- Model performance can change when the data distribution changes.
- Predictions should support analysis and learning rather than be treated as guaranteed decisions.

Responsible AI Considerations

An AI system should be transparent about uncertainty, limitations, and the nature of its training data. For financial prediction, outputs should not be presented as guaranteed investment outcomes. For sentiment analysis, privacy, dataset licensing, representativeness, and potential classification bias should be considered.

4.9 EXPECTED OUTCOMES

Project	Expected Outcome	Evaluation Focus
Stock Price Predictor	A reproducible regression pipeline capable of generating price estimates from historical features.	MAE, RMSE, R ² , chronological validation.
Sentiment Analysis	A text-classification pipeline capable of assigning positive, negative, or neutral labels.	Accuracy, precision, recall, F1, confusion matrix.

Expected Technical Outcomes

- Better understanding of supervised learning workflows.
- Practical experience with Python-based data processing.
- Ability to distinguish regression from classification tasks.
- Understanding of feature engineering for numerical and text data.
- Ability to interpret multiple evaluation metrics.
- Improved understanding of reproducible project structure.
- Experience documenting system limitations and responsible-use considerations.

Expected Academic Outcome

The combined project work is expected to strengthen the conceptual connection between classroom subjects such as programming, data structures, statistics, machine learning, and natural language processing. It also provides a structured foundation for more advanced AI/ML projects in later semesters.

4.10 CERTIFICATES OF COMPLETION AND SUPPORTING DOCUMENTS

The internship certificate supplied with the report identifies SUMIT KUMAR as an Artificial Intelligence Intern and states that the programme was conducted from 01/08/2026 to 01/09/2026. The certificate also carries the programme/AICTE identification shown in the supplied document.



Figure 4.4: Internship Offer Letter

The offer letter states that the programme was a one-month internship as Project Intern, with the designation Artificial Intelligence Intern, Pan India Hybrid mode, and reporting to assigned project head(s). It also instructs interns to complete assigned projects and training tasks through the platform and email.

4.10 SUPPORTING DOCUMENTS — CONTINUED

Project Themes Referenced in Programme Material

The supplied project-work screenshot lists multiple AI project options. Two of those options are used in this report: Stock Price Predictor and Sentiment Analysis on Twitter Data. The project descriptions specify historical-data-based stock prediction and NLP-based classification of tweets into positive, negative, or neutral categories.

Selected Project	Programme Description (Condensed)
Stock Price Predictor	Develop a model to predict stock prices from historical data; linear regression or more advanced techniques may be used.
Sentiment Analysis on Twitter Data	Build a sentiment-analysis tool to classify tweets as positive, negative, or neutral using NLP libraries such as NLTK or spaCy.

Academic Evidence and Transparency Note

The supplied files establish the internship offer and certificate and the programme's listed project themes. They do not provide independently verifiable project execution logs, datasets, source-code repositories, model metrics, supervisor reviews, or project-completion emails. Accordingly, this report does not invent such evidence. The technical sections describe a professional implementation plan and expected outcomes that can be used as the basis for an actual project demonstration.

Recommended Evidence for a Verified Technical Submission

- Source-code repository or project folder.
- Dataset/source citation and preprocessing notes.
- Notebook or execution log showing model training.
- Actual evaluation metrics produced by the implementation.
- Screenshots of the running application/notebook.
- Any genuine project-review or communication record supplied by the organization.

PROJECT SUMMARY AND CONCLUSION

This report presents a structured Artificial Intelligence internship project study covering two application areas: stock-price prediction and sentiment analysis. The first demonstrates numerical regression over historical time-series features, while the second demonstrates NLP-based multiclass classification over short text.

The report emphasizes the complete AI workflow rather than focusing only on model selection. Data validation, preprocessing, feature engineering, chronological or stratified evaluation, metric selection, responsible interpretation, and reproducibility are treated as essential parts of the solution.

The Stock Price Predictor provides a useful case study for understanding regression, lag features, rolling statistics, and time-aware evaluation. The Sentiment Analysis system provides a complementary case study for understanding text preprocessing, TF-IDF representation, classification, and class-wise evaluation.

The two projects together create a strong foundation for further work in AI/ML. Future enhancements could include stronger feature engineering, hyperparameter optimization, cross-validation strategies suited to the data, transformer-based NLP models, richer dashboards, model monitoring, and deployment through a web interface.

Most importantly, an AI project should be judged not only by the sophistication of its model but also by the quality of its data pipeline, evaluation design, documentation, and responsible communication of limitations.

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Appendix: Suggested Verification Checklist

- Run the stock project on a legitimate historical dataset before inserting any numerical results.
- Run the sentiment project on a legitimate labeled dataset before inserting classification metrics.
- Replace all expected-outcome wording with measured results only after actual execution.
- Attach genuine screenshots and communication evidence where available.
- Keep the final report consistent with the university's submission and academic-integrity requirements.